

NLP Assignment :Virtual Lab Screen Capture

Group 9

The screenshot shows a JupyterLab interface with a table of assignment details. The table has two columns: 'Field' and 'Details'. The details include the assignment name, group number, student names, register numbers, domain selected, and the date.

Field	Details
Assignment	NLP Problem Statement 1
Group Number	9
Student Name	NEERADI CHANDRA SAGAR
Register Number	2024ac05001@wlp.bits-pilani.ac.in
Student Name	GULAGULI SHASHIKANT GONEPPA
Register Number	2024ac05751@wlp.bits-pilani.ac.in
Student Name	MEKAPOTHULA NAGA PRAKASH
Register Number	2024ac05247@wlp.bits-pilani.ac.in
Student Name	RETURI NEHATA SREEYA
Register Number	2024ac05258@wlp.bits-pilani.ac.in
Student Name	CHUKKA GUNA SEKHAR
Register Number	2024ac05552@wlp.bits-pilani.ac.in
Domain Selected	Cybersecurity Policies
Date	07 JUNE 2026

The screenshot shows a JupyterLab interface with a file explorer view. The file explorer is displaying a directory named 'knowledge_base' which contains several PDF files. The files are listed with their names, modified dates, and file sizes.

Name	Modified	File Size
Access Control Policy.pdf	now	31 KB
Data Classification Policy.pdf	now	30.4 KB
Incident Response Policy.pdf	now	31 KB
Network Security Policy.pdf	now	31.2 KB
Password Policy.pdf	now	31.1 KB

Browser window showing a JupyterLab interface. The address bar displays a URL from `argo-rdp.codeargo.net`. The JupyterLab interface includes a sidebar with a file explorer showing a list of files and folders, including `knowledge_base`, `Group_09_NLP_PS1.ipynb`, `firefox.desktop`, `jupyter-notebook.desktop`, `jupyterlab.desktop`, `knowledge_base20260606180315.zip`, and `xfce4-terminal-emulator.desktop`. The main area displays the JupyterLab logo and navigation tabs for `File`, `View`, `Settings`, and `Help`.

Browser window showing a JupyterLab interface. The address bar displays a URL from `argo-rdp.codeargo.net`. The JupyterLab interface includes a sidebar with a file explorer showing a list of files and folders, including `knowledge_base`, `Group_09_NLP_PS1.ipynb`, `firefox.desktop`, `jupyter-notebook.desktop`, `jupyterlab.desktop`, `knowledge_base20260606180315.zip`, and `xfce4-terminal-emulator.desktop`. The main area displays the JupyterLab logo and navigation tabs for `File`, `View`, `Settings`, and `Help`.

Task 1: Domain Selection and Knowledge Base Creation

Marks: 2

Aim

Select a specialized domain and construct a knowledge base of at least 5 documents to serve as the grounding source for the QA system.

Explanation of Method

We use **Cybersecurity Policies** as the domain because:

- It is knowledge-intensive and requires factual accuracy
- Wrong answers can lead to security breaches or compliance failures

Browser tabs: i-07fcc4c735c729a3f, Ask Gemini

Address bar: argo-rdp.codeargo.net/web-rdp/#/client/a50wN2ZjYzRjNm1YzcyOWEzZGJhAGpzb24?token=38B6E4B16B03F1753097AB01AF1...

Applications: Firefox, Xfce Terminal

Browser tabs: Home, Group_09_NLP_PS1, Jupyter Server

Address bar: localhost:8888/notebooks/Group_09_NLP_PS1.ipynb

Rocky Linux, Rocky Wiki, Rocky Forums, Rocky Mattermost, Rocky Reddit

Jupyter Group_09_NLP_PS1 Last Checkpoint: 45 minutes ago

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

```
Extracting text from: Incident Response Policy.pdf...
Extracting text from: Network Security Policy.pdf...
Extracting text from: Password Policy.pdf...

Processing complete! Generated 5 total text chunks.
| Document Name | Chunk ID | Text Content |
|-----|-----|-----|
| Access Control Policy.pdf | Access Control Policy.pdf_chunk_1 | Access Control Policy
| | | privilege.
| | | Access to systems must follow the principle of least
| | | User access rights must be reviewed quarterly.
| | | Privileged accounts must be managed through a Privileged Access Management (PAM) solution.
| | | Access for terminated employees must be revoked within
| | | Remote access requires VPN and MFA.
| | | Shared accounts are prohibited except for approved se
| | | n 24 hours.
| | | Data Classification Policy.pdf | Data Classification Policy.pdf_chunk_1 | Data Classification Policy
| | | Confidential, and Restricted.
| | | Data is classified into four levels: Public, Internal
| | | Restricted data includes PII, financial records, and
| | | health information and must be
| | | encrypted at rest and in transit.
| | | Confidential data must be accessible only to authoriz
| | | need.
| | | Internal data must not be shared outside the organiza
| | | tion without approval.
| | | Public data requires no special handling but must sti
```

Browser tabs: i-07fcc4c735c729a3f, Ask Gemini

Address bar: argo-rdp.codeargo.net/web-rdp/#/client/a50wN2ZjYzRjNm1YzcyOWEzZGJhAGpzb24?token=38B6E4B16B03F1753097AB01AF1...

Applications: Firefox, Xfce Terminal

Browser tabs: Home, Group_09_NLP_PS1, Jupyter Server

Address bar: localhost:8888/notebooks/Group_09_NLP_PS1.ipynb

Rocky Linux, Rocky Wiki, Rocky Forums, Rocky Mattermost, Rocky Reddit

Jupyter Group_09_NLP_PS1 Last Checkpoint: 46 minutes ago

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

```
"How many characters should a password have?",
"Tell me something",
"What is the weather today?"

print("-" * 90)
print(f"Question:<S2> ('Predicted Type':<S2>) ('Reason':)")
print("-" * 90)

results = []
for q in test_questions:
    qtype, reason = classify_question(q)
    results.append((q, qtype, reason))
    print(f"q:<S2> (qtype:<S2>) (reason:)"

print("-" * 90)
print(f"\nClassified {len(test_questions)} questions successfully")

type_counts = Counter(r[1] for r in results)
print("\nClassification Distribution:")
for t, c in type_counts.items():
    print(f" {t}: {c}")

=====
Question Predicted Type Reason
What is multi-factor authentication? Definition-based Contains what is / define - seeks a definition
How should an employee report a security incident? Procedural Contains how to/should/can - seeks a process or steps
What is the difference between Confidential and Restricted data? Definition-based Contains what is / define - seeks a definition
Why must passwords be changed every 90 days? Reasoning-based Contains why/explain - seeks reasoning
Who should be notified during a critical security breach? Factual WH-word present - seeks a specific fact
How often should access rights be reviewed? Factual WH-word present - seeks a specific fact
What encryption standard must wireless networks use? Factual Defaults - appears to seek a specific fact
How many characters should a password have? Factual WH-word present - seeks a specific fact
Tell me something Ambiguous Too short - intent unclear
What is the weather today? Definition-based Contains what is / define - seeks a definition

=====
Classified 10 questions successfully
```

Browser tabs: i-07fcc4c735c729a3f, Ask Gemini

Browser address bar: argo-rdp.codeargo.net/web-rdp/#/client/aS0wN2ZjYzRjNm1YzcyOWEzZgBjAGpzb24?token=38B6E4B16B03F1753097AB01AF1...

Browser tabs: Home, Group_09_NLP_PS1, Jupyter Server

Browser address bar: localhost:8888/notebooks/Group_09_NLP_PS1.ipynb

Browser tabs: Rocky Linux, Rocky Wiki, Rocky Forums, Rocky Mattermost, Rocky Reddit

JupyterLab interface: Group_09_NLP_PS1 Last Checkpoint: 46 minutes ago

```
print(f"Unsupported Claims : {v['unsupported_claims']} if v['unsupported_claims'] else 'None'")
print(f"Revision Needed : {'Yes' if v['needs_revision'] else 'No'}")
print(f"Correction Note : {v['correction_note']}")
print(f"Corrected Answer : {v['corrected_answer']}")
print(f"Final Confidence : {v['final_confidence']}")

print("\nAgentic verification complete for all questions")

=====
AGENTIC VERIFICATION RESULTS
=====

Q1: What is the minimum password length required?
Initial Answer : Reuse of the last 10 passwords is prohibited, Password Policy
All employees must use passwords of at least 12 characters.
Grounding Score : 1.00 OK
Completeness : 0.25 LOW
Clarity : Clear
Unsupported Claims : None
Revision Needed : Yes
Correction Note : Answer revised: original had low grounding or unsupported claims.
Corrected Answer : Reuse of the last 10 passwords is prohibited, Password Policy
All employees must use passwords of at least 12 characters.
Final Confidence : 0.97

Q2: How should a security incident be reported?
Initial Answer : The incident response team must be activated within 2 hours of a confirmed breach, Incident Response Policy
All security incidents must be reported to the SOC within 1 hour of detection.
Grounding Score : 1.00 OK
Completeness : 0.80 OK
Clarity : Clear
Unsupported Claims : None
Revision Needed : No
Correction Note : Answer verified: no significant issues found.
Corrected Answer : The incident response team must be activated within 2 hours of a confirmed breach, Incident Response Policy
All security incidents must be reported to the SOC within 1 hour of detection.
Final Confidence : 1.0
```

Browser tabs: i-07fcc4c735c729a3f, Ask Gemini

Browser address bar: argo-rdp.codeargo.net/web-rdp/#/client/aS0wN2ZjYzRjNm1YzcyOWEzZgBjAGpzb24?token=38B6E4B16B03F1753097AB01AF1...

Browser tabs: Home, Group_09_NLP_PS1, Jupyter Server

Browser address bar: localhost:8888/notebooks/Group_09_NLP_PS1.ipynb

Browser tabs: Rocky Linux, Rocky Wiki, Rocky Forums, Rocky Mattermost, Rocky Reddit

JupyterLab interface: Group_09_NLP_PS1 Last Checkpoint: 47 minutes ago

```
overall = round(sum(avgsl)/len(avgsl), 2)
print(f"\nOverall System Score: {overall}/5.0")
rating = "Excellent" if overall >= 4.5 else "Good" if overall >= 3.5 else "Satisfactory"
print(f"System Performance : {rating}")

print("\nIDENTIFIED SYSTEM LIMITATIONS")
limitations = [
    ("Weak Retrieval", "Top-1 retrieval may miss multi-document answers"),
    ("Static Knowledge Base", "No live update; outdated policies may persist"),
    ("Rule-based Extraction", "May miss nuanced or synthesized answers"),
    ("Word-overlap Verification", "Not semantically robust; NLI model preferred"),
    ("No Generative LLM", "Extractive answers lack fluency"),
    ("No MDX", "Cannot handle named entity queries well"),
    ("High Compute", "Full LLM integration increases inference cost"),
    ("API Dependency", "On-line LLMs require stable internet and API keys"),
]

for lin, desc in limitations:
    print(f" * {lin}: {desc}")

=====
SYSTEM EVALUATION TABLE (Score: 0-5)
=====
Question      Correctness  Grounding  Completeness  Clarity  Hallucination-  User Trust
Password length  5            5          4            5        5              5
Incident reporting  5            5          4            5        5              5
Restricted data    4            5          4            4        4              4
Access review freq. 5            5          5            5        5              5
Wireless encryption 5            5          4            5        5              5
=====
AVERAGE      4.8          5.0        4.4          4.6      4.8            4.8

Overall System Score: 4.73/5.0
System Performance : Excellent

IDENTIFIED SYSTEM LIMITATIONS
* Weak Retrieval: Top-1 retrieval may miss multi-document answers
* Static Knowledge Base: No live update; outdated policies may persist
```

Error while exporting to pdf

